北京郵電大學

Queen Mary University of London

For examiners' use only

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Total	

EBU6304 A

Joint Programme Examinations 2023/24

EBU6304 Software Engineering

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Ling Ma, Dr Gokop Goteng, Dr Ethan Lau, Dr Alan Wong

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

You are leading a software development team to build a new fitness app for a startup company. The project aims to develop an interactive, user-friendly app that can provide users with guided workout routines, track activity, and offer personalised recommendations. The startup has ambitious growth plans and expects frequent customer feedback and updates to enhance features and accommodate changing market demands. Your team members are highly skilled and experienced developers. Based on this brief description of the scenario, answer the following questions.

a) Discuss why Agile is a suitable software process for this project. Provide specific reasons by linking the scenario to Agile principles.

[8 marks]

	Do not write in this column
	8 marks

b) Identify a functional requirement that has the highest priority and provide your reasons why it has the highest priority. Write a detailed user story for that function.

[8 marks]

	Do not write in this column
	8 marks

c) Identify a non-functional requirement and rewrite it in a way that can be verified.

[4 marks]

	Do not write in this column
	4 marks

d) You need to test a program in the App that calculates the Body Mass Index (BMI), which is a measure that estimates whether the user's weight is healthy relative to their height. The BMI calculation divides the user's weight in kilograms (kg) by their height in meters (m) squared. Mathematically, it's expressed as: $BMI = \frac{weight\ (kg)}{height(m)^2}$

You are going to use Test-Driven Development and test this program using JUnit. Write the test code in Java with the two test cases. 1) height = 1.8m, weight = 75kg, BMI=23; 2) height = 1.6m, weight = 55kg, BMI=21.5.

[5 marks]

	Do not write in this column
	5 marks

Question marking: $\frac{}{8} + \frac{}{8} + \frac{}{4} + \frac{}{5} = \frac{}{25}$

Question 2

a) A hospital aims to streamline its operations by developing a medical record management system for automation. The system is intended for use by doctors, nurses, pharmacists, and patients. Each patient possesses a medical card containing the patient's name and a unique medical identification number. To visit the hospital, a patient must schedule an appointment with a doctor.

During the visit, the doctor accesses the patient's record using their unique identification number and issues prescriptions. These prescriptions are automatically transmitted to the pharmacists and nurses through a monitoring system, facilitating further diagnosis on the patient. Out-patients can return home after receiving medications from the pharmacist and nurse. On the other hand, In-patients are escorted to the hospital ward, where they stay under the nurse's care until the doctor discharges them to go home upon recovery.

Use the information above to answer the following questions:

[13 marks]

i) Describe the three main stereotypes of classes for analysis. Identify the classes under each category from the scenario.

(7 marks)

ii) Create a simple UML class diagram for the hospital management system that illustrates the relationships among classes, including inheritance relationships. (Include only class names in the diagram.)

(6 marks)

	Do not write in this column

b) Which two fundamental design concepts are used to ensure less dependencies within the system? Explain your answers.

[6 marks]

	Do not write in this column
	6 marks

c) A software component is used to classify student exam results into grades of A (70%-100%), B (60%-69%), C (50%-59%), D (45%-49%), E (40%-44%), and F (0%-39%). Describe how Partition Testing can be used to test this component. (You may include some illustration or a diagram to explain your answer).

[6 marks]

	Do not write in this column	
		6 marks

Question marking: $\frac{}{13} + \frac{}{6} + \frac{}{6} = \frac{}{25}$

Question 3

a) List five most significant features of using RESTful architecture for development compared to other approaches.

[5 marks]

	Do not write in this column
	5 marks

b) FlexiSoft Company plans to develop a next generation video editing software – FlexiVid Editor and initiates the project management planning process to ensure the product is in accordance with the quality standards. Answer the following questions:

[15 marks]

i) Table 1 below defines the activities involved for the FlexiVid Editor project. Draw an Activity-on-node (AON) diagram and correctly label the early start (ES), late start (LS), early finish (EF) and late finish (LF) for each nodes.

Activity	Description	Duration (weeks)	Immediate predecessor activities
A	Define requirement	2	-
B	Assign team	3	-
C	Selecting video codecs	4	A
D	Code software	3	B, A
E	Build software	8	D, C
F	Test software	3	C
G	Audit software	2	E
H	Deploy software	3	F, G

Table 1: Project activities.

(8 marks)

ii) Identify the critical path chain and the maximum allowable duration in completing the project.

(3 marks)

iii) Identify the non-critical time for each activity and the total non-critical time taken to complete the project without increasing the overall project completion time.

(4 marks)

	Do not write in this column

c) A business start-up company uses a `PaymentProcessor` class to process payment. Currently, it only supports debit card payments, as shown in Figure 1 below. However, as the business grows, it needs to extend the application to support Alipay payments. To achieve this, the developer has suggested modifying the existing `PaymentProcessor` class and adding a new function `processAlipayPayment()` as the new payment method. Explain one design principle in this case that the developer violates. Also, suggest a way that the developer can improve the existing `PaymentProcessor` class to achieve this goal.

```
class PaymentProcessor {
    void processDebitCardPayment() {
        // function code to process debit card payment
    }
}
```

Figure 1. PaymentProcess class

[5 marks]

	Do not write in this column
	5 marks

Question marking: $\frac{}{5} + \frac{}{15} + \frac{}{5} = \frac{}{25}$

Question 4

A technology company requires a software product for managing its graduate recruitment and associated management-training programme. Answer the following questions.

a) Describe briefly the company's business risk, if it were to:
   i) Develop the product 'in-house'.
   ii) Purchase the product from a proprietary software supplier.
   iii) Use an Open Source Software (OSS) product.

[6 marks]

	Do not write in this column
	6 marks

b) The company decides to develop the product in-house, to tailor to its specific requirements. Name some of the important quality attributes that the company might want to see in this software, and explain whether it might be possible to include all these attributes in the product.

[8 marks]

	Do not write in this column
	8 marks

c) How might a version control system help the company manage the quality of the software that is being implemented?

[4 marks]

	Do not write in this column

		4 marks

d) Briefly describe the concept of disparate impact, and explain whether this is justified when developing 'fair' software or products. Give one example to illustrate your answer.

[7 marks]

	Do not write in this column
	7 marks

Question marking: $\frac{}{6} + \frac{}{8} + \frac{}{4} + \frac{}{7} = \frac{}{25}$

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2023-2024

Rough Working

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